

A Comparative Analysis of YOLO Models on Human Detection for Drone-Based Search and Rescue

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Abstract—Unmanned Aerial Vehicles (UAVs), or drones, are increasingly vital in Search and Rescue (SAR) operations, offering rapid coverage of challenging terrains. Effective human detection in drone-captured imagery is crucial for mission success but faces obstacles such as varied environmental conditions and real-time processing demands. This paper presents a comprehensive comparative analysis of YOLOv8 and YOLOv11 model variants (specifically 'n' and 's' versions) for human detection using the specialized Search and Rescue Dataset (SARD). The models were trained with minimal preprocessing and no explicit data augmentation, focusing on their core performance with pre-trained weights and a consistent training regimen (20 epochs, 640x640 image size). Performance was evaluated based on standard metrics including Precision, Recall, mean Average Precision (mAP-50 and mAP50-95), and inference time. Results indicate that YOLOv8s achieved the highest detection accuracy with a mAP50-95 of 0.606, outperforming YOLOv8n (0.591), YOLOv11s (0.585), and YOLOv11n (0.575). In terms of speed, YOLOv8n was the fastest, with an inference time of 7.83 ms, followed by YOLOv11n (9.89 ms), YOLOv8s (12.18 ms), and YOLOv11s (15.5 ms). This study is expected to provide information on the capabilities of YOLO variants, especially YOLOv8 and YOLOv11, by highlighting that YOLOv8 offers the best balance of accuracy in this experimental setting. In addition, YOLOv8 excels in terms of speed, making it a model of choice for practical drone-based SAR applications.

Keywords—Computer Vision, Deep Learning, Human Detection, Image Detection, Search and Rescue (SAR), YOLO

I. INTRODUCTION

The effectiveness of emergency response in search and rescue (SAR) operations is highly dependent on rapid and precise location detection, especially in environments that are inaccessible or hazardous to rescue teams. In highly time-sensitive operations, the ability to quickly identify human subjects in aerial imagery is critical to improving survival outcomes and increasing operational efficiency [1], [2]. To assist SAR operations, currently Unmanned Aerial Vehicles (UAVs) are increasingly being used, also known as drones, because they can cover large and complex areas from the air quickly [3]. However, even today it is still very difficult to accurately identify humans in images taken by drones. This is

due to factors such as weather conditions and slow real-time processing speeds [4].

More researchers are using object detection algorithms that use deep learning to solve this problem. The You Only Look Once (YOLO) algorithm has become very popular because it can process data in real time and find things accurately and has proven to be robust in various tasks [5]. The YOLO model is renowned for its ability to balance detection speed and accuracy with minimal trade-off, making it ideal for use in real-time applications such as search and rescue operations [6].

Previous studies have observed different types of YOLO models by comparing their performance mainly on public datasets involving common objects or generic scenarios [7]. However, limited research has specifically analyzed and compared the efficacy of the latest YOLO versions, particularly YOLOv8 and YOLOv11, in detecting humans in drone imagery under SAR contexts. Consequently, there exists a notable research gap concerning the comparative performance analysis of these advanced YOLO architectures specifically for SAR applications.

To bridge this research gap, the present study aims to conduct a comprehensive comparative analysis of YOLOv8 and YOLOv11. Our solution focuses explicitly on evaluating these two models using a dedicated drone image dataset captured during realistic SAR scenarios, analyzing performance metrics including detection accuracy and processing speed. Through this comparative approach, we aim to provide clear insights into the capabilities, strengths, and limitations of these YOLO models.

Our motivation for selecting YOLO as our primary methodological framework lies in its proven efficiency and robust performance in real-time detection tasks. Specifically, YOLOv8 and YOLOv11 represent some of the latest advancements within this family, offering improvements in detection accuracy and computational speed [8]. Given the critical requirement for timely and accurate human detection during rescue operations, exploring and benchmarking these latest versions is both academically valuable and practically significant.

This research addresses the identified gap by systematically evaluating YOLOv8 and YOLOv11, providing valuable benchmarks and insights for deploying drone-assisted SAR missions. The outcomes of this study are expected to guide both future academic research and practical implementations in the field of aerial human detection for emergency scenarios.

II. RELATED WORKS

The You Only Look Once (YOLO) family of object detection algorithms has been a subject of extensive research and development due to its compelling balance of speed and accuracy, making it highly suitable for real-time applications. Nazir et al. conducted a review encompassing YOLO versions from YOLOv2 to YOLOv7. Their work highlighted the popularity of YOLO for its efficiency and pointed towards future research in enhancing robustness to environmental conditions [9]. Similarly, Mulajkar provided a comprehensive review covering YOLO versions v1 through v8, detailing their architectures, the datasets commonly used for evaluation, achieved mAP scores, and significant architectural shifts over the years [10]. Dhruthi L. et al. also compared YOLO variants from YOLOv3 to YOLOv7 on the MS COCO dataset, assessing their performance on both CPU and GPU platforms to guide users in model selection based on accuracy and computational efficiency [11].

The application of YOLO models for human detection in drone imagery, particularly for Search and Rescue (SAR) operations, has garnered significant attention. Hafizhelmi et al. investigated the use of YOLOv5 for human detection from drones in SAR scenarios, proposing that employing a 6x zoom magnification could significantly improve victim localization. Their results showed that while YOLOv5 performed reasonably (mAP50 of 0.346), the zoom magnification boosted mAP50 to 0.597, though they noted room for improvement in reducing false detections [12]. Valarmathi et al. focused on both human detection and action recognition in disaster scenarios using drones, employing a YOLOv3 algorithm. Their proposed method achieved a high accuracy of 94.9% for these combined tasks and processed images in just 0.40 milliseconds, outperforming other contemporary models like F-RCNN and SSD [13].

More research conducted by Chebtha et al. compared the accuracy and speed of various YOLO versions for real-time drone detection, identifying YOLOv5 and YOLOv8 as particularly effective. Their experiments showed YOLOv8x achieving a 98% mAP with an inference time of 31 FPS, outperforming YOLOv6s (90.8% mAP, 70 FPS) [14]. Mathew et al. compared YOLOv2 and YOLOv3 for object detection from aerial images using the DOTA dataset. While YOLOv2 achieved a higher F1-score for specific classes like aeroplanes (99%) and swimming pools (83%), YOLOv3 demonstrated superior speed with an average detection time of 2.5 seconds [15].

Furthermore, the application of YOLO models extends to specialized imagery, such as thermal images. Rizk et al. evaluated the performance of YOLOv8 for human detection specifically in thermal images for SAR. They utilized a novel dataset of 17,148 grayscale thermal images and reported that YOLOv8 demonstrated a high average precision of 95% in human detection [16]. The collective body of these related works establishes the state-of-the-art in YOLO-based object detection and its application to SAR, highlighting a

continuous drive for improved accuracy and efficiency. Our research builds upon this foundation by providing a focused, up-to-date comparison of YOLOv8 and YOLOv11, aiming to guide practitioners and researchers in selecting the most suitable models for the critical task of human detection in drone-assisted SAR operations.

III. METHODOLOGY

To comprehensively compare YOLOv8 and YOLOv11 for human detection in drone-based SAR, a structured methodology is proposed, encompassing dataset selection, data preprocessing, model training, and rigorous evaluation.

A. Dataset

This research employs the Search and Rescue Dataset (SARD), explicitly designed to support the development and evaluation of artificial intelligence models for locating, identifying, and tracking individuals in distress during disaster or emergency scenarios. This dataset includes images and labels collected from real-world drone imagery. The drone used for capturing these images was equipped with a high-resolution camera, specifically the DJI Phantom 4A. Table 1 below outlines the structure of the dataset, including the distribution of images across training, testing, and validation sets:

TABLE I. DATASET DISTRIBUTION

Dataset Partition	Number of Images
Training Set	4,041
Validation Set	1,144
Testing Set	570

Overall, the use of such a specialized dataset is crucial in this research as it allows for robust evaluation of YOLOv8 and YOLOv11 in realistic, practical scenarios. Given the complexity and the critical nature of SAR operations, datasets that accurately reflect real-world challenges and conditions are essential for drawing meaningful conclusions about algorithm performance and practical applicability.

B. Proposed Method

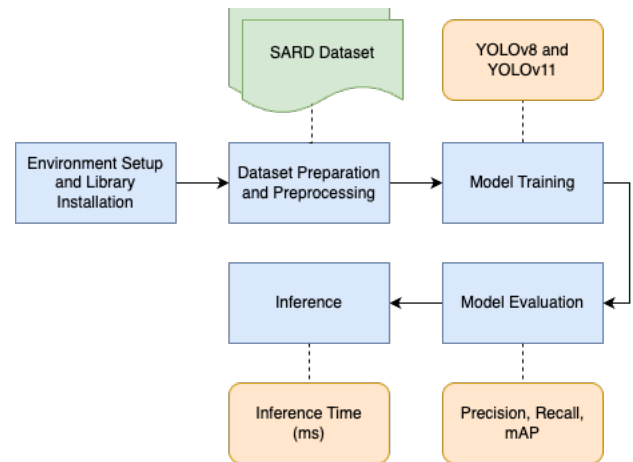


Fig. 1. Proposed method workflow.

The proposed method starts with dataset preparation, where the SARD dataset is imported to the program, and the data path is properly configured for model training. Image

preprocessing is conducted, including resizing and normalization, ensuring images conform to the input specifications required by YOLO models.

Training involves employing YOLOv8 and YOLOv11 separately, using Ultralytics' pre-trained weights. Model configurations are carefully selected, with hyperparameters such as epochs set to 20, image size to 640, and batch size to 16. Each model's training process is closely monitored to optimize performance and prevent overfitting.

The trained models are then rigorously evaluated on the validation set to measure performance using metrics such as mean Average Precision (mAP). Model inference time is also recorded to assess the practical applicability of each YOLO variant. Finally, the inference process is visualized, enabling qualitative assessment of model performance, highlighting detection accuracy, and practical inference speeds, essential for real-time SAR missions.

C. Data Preprocessing

For this experiment, data preprocessing was kept to a minimum, primarily relying on the standard procedures handled by the Ultralytics YOLO framework. The key aspects included ensuring Annotation Formatting was consistent with YOLO requirements. This typically involves separate label file per image, where each line specifies the object's class ID and its normalized bounding box coordinates. We use yaml file, which directed the model to the pre-split training, validation, and test image directories and contained class information. Image Resizing to the specified input size (imgsz=640) is a standard step handled internally by the YOLO model during training to ensure consistent input dimensions. Similarly, normalization of pixel values, is an automatic process within modern YOLO implementations.

The experiment focused on training the models with the dataset in its near-original state after basic formatting. As previously noted, the SARD dataset was used with its existing pre-defined splits for training, validation, and testing, eliminating the need for manual dataset partitioning.

D. Model Training

We utilized YOLOv8 and YOLOv11 in this research. While YOLOv8 established a new benchmark in real-time object detection with its refined architecture featuring an improved CSPDarknet backbone and an anchor-free, decoupled head, subsequent iterations have continued to push the boundaries of performance and efficiency. An advancement in this line is a model that has been designated as YOLOv11, which introduces significant architectural modifications aimed at superior computational efficiency and accuracy. Unlike the foundational design of YOLOv8, YOLOv11 is reported to incorporate novel components such as a C3k2 block and advanced spatial attention mechanisms. These changes reportedly result in a model that not only achieves a higher mean average precision (mAP) but does so with a considerable reduction in parameters and floating-point operations (FLOPs). Consequently, the primary distinction lies in YOLOv11's enhanced efficiency, offering faster inference speeds, particularly on CPU hardware, and a more lightweight architecture without compromising the robust detection capabilities that characterized its predecessors [17], [18].

Both YOLOv8 and YOLOv11 models will be trained on the SARD dataset using the pre-defined splits. We utilized

the 'n' and 's' version for both YOLOv8 and YOLOv11 models. It is important to note that the suffixes 'n' (nano) and 's' (small) denote the model's scale. Nano models are the lightest and fastest with fewer parameters, while small models are slightly larger, aiming for higher accuracy at the cost of some speed. Table 2 shows the difference of each model versions.

TABLE II. MODEL VERSIONS DIFFERENCE

Model	Parameters (Million)
YOLOv8n	3.15
YOLOv8s	11.16
YOLOv11n	2.62
YOLOv11s	9.44

The models used in this study are divided into two scales, each marked with the suffix 'n' (nano) and 's' (small). Both have differences mainly in terms of size. The 'nano' version of the YOLOv8n and YOLOv11n models is designed for maximum efficiency. YOLOv8n has 3.15 million parameters and YOLOv11n has 2.62 million parameters. The number of parameters for both is much smaller than the 'small' version. The model becomes lighter and optimized for speed, making it suitable for real-time applications or implementation on hardware with limited resources. In contrast, in the 'small' version, YOLOv8 has 11.16 million parameters and YOLOv11 has 9.44 million parameters. So the capacity of these two variant models allows for higher learning potential and thus higher accuracy, although at the expense of slower inference speed and higher computation.

The training process in the study utilized the Transfer Learning technique, where the model was initialized with weights that had been previously trained on a large benchmark dataset such as the COCO dataset. Key Training Parameters, as indicated by the experimental setup, include 20 epochs, an image size of 640x640 pixels, and a batch size of 16. All training will be conducted on GPU to manage the computational demands. The performance on the pre-defined Validation set will be monitored throughout the training process. This continuous validation guides hyperparameter tuning and allows for the implementation of early stopping, if necessary, thereby preventing overfitting and ensuring the model generalizes well to unseen data.

E. Evaluation Metrics

The performance of the trained YOLOv8 and YOLOv11 models will be quantitatively assessed using a suite of standard object detection metrics. The Intersection over Union (IoU) is fundamental, measuring the overlap between predicted and ground-truth bounding boxes. This forms the basis for classifying detections as True Positives (TP), False Positives (FP), or False Negatives (FN). Based on these classifications, we use precision, indicating the accuracy of positive predictions and thus minimizing false alarms. Recall is also used to measure the model's ability to find all relevant objects and thereby minimizing missed detections.

The Average Precision (AP) provides a single summary statistic for performance on a specific class, calculated as the area under the Precision-Recall curve. For this study, the Mean Average Precision (mAP) will be the primary metric, representing the average of AP values (or the AP for the single

"human" class in this context). This study reports two common mAP variants, namely mAP50 and mAP50-95. mAP50 represents the mean Average Precision calculated at a single, fixed Intersection over Union (IoU) threshold of 0.50. It measures the model's ability to correctly detect objects with at least 50% overlap with the ground-truth bounding box. Meanwhile mAP50-95 provides a more comprehensive and stringent evaluation of localization accuracy. It is calculated by averaging the mAP value across ten different IoU thresholds, starting from 0.50 up to 0.95 with a step size of 0.05. This rewards models that are accurate over a wide range of overlap requirements.

IV. RESULTS AND DISCUSSION

A. Performance Evaluation

Model performance is measured on a predefined validation set with the evaluation metrics described earlier. This can be seen in Table 3. We compare four variants of the YOLO model: YOLOv8n, YOLOv8s, YOLOv11n, and YOLOv11s. In the YOLOv8 version, the YOLOv8s variant consistently outperforms the YOLOv8n variant across all evaluation metrics. Specifically, YOLOv8s achieves a higher Precision (0.941 vs. 0.933), Recall (0.865 vs. 0.85), mAP-50 (0.931 vs. 0.917), and notably, a higher mAP50-95 (0.606 vs. 0.591). This result aligns with general expectations where larger model variants (like 's' for small, compared to 'n' for nano) typically possess more parameters, enabling them to learn more complex features and thus achieve higher accuracy.

TABLE III. MODEL'S PERFORMANCE

Model	Precision	Recall	mAP-50	mAP50-95
YOLOv8n	0.933	0.85	0.917	0.591
YOLOv8s	0.941	0.865	0.931	0.606
YOLOv11n	0.908	0.857	0.916	0.575
YOLOv11s	0.925	0.847	0.914	0.585

Analyzing the YOLOv11 family, the YOLOv11s variant shows higher Precision (0.925 vs. 0.908) and a better mAP50-95 score (0.585 vs. 0.575), suggesting it is more accurate in its positive predictions and performs better at stricter localization criteria. However, the YOLOv11n model records a slightly higher Recall (0.857 vs. 0.847) and a marginally better mAP-50 (0.916 vs. 0.914). This mixed outcome suggests that while the larger YOLOv11s model excels in precision and high-IoU performance, the lighter YOLOv11n variant might be more effective at identifying a slightly larger proportion of all relevant objects and performs competitively at IoU threshold of 0.5.

When comparing the top-performing models from each family, YOLOv8s surpasses YOLOv11s in Precision (0.941 vs. 0.925), Recall (0.865 vs. 0.847), mAP-50 (0.931 vs. 0.914), and mAP50-95 (0.606 vs. 0.585). This outperformance by YOLOv8s is significant, as newer iterations like YOLOv11 are generally developed with architectural enhancements. The superior performance of YOLOv8s in this instance could be attributed to various factors, including the specific characteristics of the SARD dataset aligning more favorably with the YOLOv8 architecture, or the particular training parameters used in the

experiment potentially optimizing YOLOv8s more effectively. It's also possible that the specific variants of YOLOv8 and YOLOv11 being compared play a role, as performance can vary significantly across model sizes.

From a practical SAR perspective, evaluation metrics translate directly to operational effectiveness. The results of mAP-50 indicates that if a human is present, the models are very likely to detect them with at least a 50% accurate bounding box overlap. For a rescue team, this means the system can reliably reduce a vast search area to a few high-probability locations, saving critical time. A high mAP-50 minimizes the chances of completely missing a victim (False Negatives).

Meanwhile mAP50-95 metric provides insight into the model's localization precision. The score of 0.606 achieved by YOLOv8s suggests that while the model is excellent at identifying the presence of a person, its ability to draw a pixel-perfect bounding box is moderate. In a real-world SAR mission, this is often an acceptable trade-off. A slightly imprecise bounding box that correctly locates a victim is far more valuable than no detection at all. The superior mAP50-95 score of YOLOv8s over its counterparts implies that in complex scenarios where a victim might be partially obscured or in an unusual pose, it would provide a more reliable and accurately placed marker for rescue personnel to investigate, thereby increasing the overall efficiency and success probability of the mission.

B. Inference Time Analysis

The inference time results presented in Table 3, derived from processing 100 random validation images, provide crucial insights into the computational efficiency of the four YOLO model variants: YOLOv8n, YOLOv8s, YOLOv11n, and YOLOv11s. Within the YOLOv8 family, the nano variant, YOLOv8n, is significantly faster, clocking in at 7.83 ms, compared to the YOLOv8s variant at 12.18 ms. This aligns with the general principle that smaller model architectures ('n' for nano versus 's' for small) have fewer parameters and computational demands, leading to quicker inference. A similar trend is observed within the YOLOv11 family, where YOLOv11n (9.89 ms) outperforms its larger counterpart, YOLOv11s (15.5 ms). Overall, YOLOv8n emerges as the fastest model in this comparison, while YOLOv11s is the slowest, underscoring the inherent trade-off between model size/complexity and processing speed.



Fig. 2. Model inference results.

TABLE IV. MODEL'S INFERENCE TIME

Model	Inference Time (ms)
YOLOv8n	7.83
YOLOv8s	12.18
YOLOv11n	9.89
YOLOv11s	15.5

When compared across different YOLO versions of the same variant, the YOLOv8 model shows a speed advantage over the YOLOv11 model in this experiment. Specifically, YOLOv8n takes an inference time of 7.83 ms, which is faster than YOLOv11n which takes an inference time of 9.89 ms, and YOLOv8s takes an inference time of 12.18 ms, which is faster than YOLOv11s which takes an inference time of 15.5 ms. This suggests that specifically for this SARD dataset, the YOLOv8 architecture is more optimal for inference speed. While later versions such as YOLOv11 introduce architectural enhancements, which aim to improve accuracy and efficiency. These results suggest that this particular YOLOv8 variant is more performant in terms of latency. It is important to note that inference time is affected by many factors, including the complexity of the model architecture, the specific hardware used (e.g., GPU type), the batch size during inference, and the underlying software optimizations or library versions.

The implications of these inference times are particularly significant for real-world applications like drone-based Search and Rescue (SAR), where real-time processing is paramount for timely decision-making and effective intervention. YOLOv8n's rapid inference time of 7.83 ms translates to a potential processing capability of approximately 127 frames per second (FPS), making it an excellent candidate if its detection accuracy meets the mission requirements. Even YOLOv8s, at 12.18 ms (~82 FPS), offers substantial real-time performance. While YOLOv11s is the slowest at 15.5 ms (~64 FPS), it might be considered if it offered a compelling accuracy advantage, though previous data (Table II from the prior turn) indicated YOLOv8s was more accurate in this specific experimental context. Ultimately, the selection of an optimal model for deployment, especially on resource-constrained platforms like drones, necessitates a careful balance of inference speed, detection accuracy, and the specific operational constraints of the SAR mission. The visual examples in Fig. 2 demonstrate the models' capability to detect humans, but the speed at which these detections are generated, as quantified in Table 3, is a critical factor for their practical utility.

V. CONCLUSIONS

This study aimed to provide a comprehensive comparative analysis of YOLOv8 and YOLOv11 variants (specifically 'n' and 's' versions) for the critical task of human detection in drone-based Search and Rescue (SAR) operations, utilizing the SARD dataset. The evaluation focused on key performance metrics including Precision, Recall, mAP-50, mAP50-95, and inference time, with models trained under consistent conditions and minimal preprocessing.

The experimental results revealed that the YOLOv8s model demonstrated the highest overall detection accuracy, achieving a Precision of 0.941, Recall of 0.865, mAP-50 of 0.931, and a mAP50-95 of 0.606. This surpassed the performance of YOLOv8n (mAP50-95: 0.591), YOLOv11s

(mAP50-95: 0.585), and YOLOv11n (mAP50-95: 0.575). In terms of computational efficiency, YOLOv8n was the fastest model with an inference time of 7.83 ms. This was followed by YOLOv11n at 9.89 ms, YOLOv8s at 12.18 ms, and YOLOv11s being the slowest at 15.5 ms. These findings suggest that within this specific experimental context, the YOLOv8s model offered the best balance of high accuracy, while the YOLOv8n variant provided the optimal processing speed. Notably, the YOLOv8 variants generally outperformed their YOLOv11 counterparts in both accuracy and speed.

This study has certain limitations as the findings are based on the SARD dataset, and performance may vary on datasets with different characteristics or in diverse SAR environments. Furthermore, inference times were measured on specific hardware, and real-world deployment on drone-embedded systems could yield different performance profiles.

Future research should explore the impact of more extensive data augmentation techniques tailored to aerial imagery challenges. Evaluating these models with longer training periods, comprehensive hyperparameter optimization, and on a wider variety of SAR datasets representing different terrains and conditions would provide more generalizable insights. Testing on actual drone hardware is also crucial to assess practical inference speeds and power consumption. Additionally, investigating the performance of other model sizes within the YOLOv8 and YOLOv11 families, or newer YOLO versions as they become available, will continue to advance the field.

In conclusion, this paper contributes valuable benchmarks for YOLOv8 and YOLOv11 variants in the context of human detection for drone-based SAR. It underscores the critical trade-off between detection accuracy and inference speed, providing a practical guide for researchers and practitioners in selecting appropriate deep learning models to enhance the effectiveness of life-saving SAR operations.

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The Search and Rescue Dataset (SARD) used to support the findings of this study is publicly available and can be accessed at this link: <https://www.kaggle.com/datasets/nikolasgegenava/sard-search-and-rescue/data>.

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